

## Direct Kinematic Model

|   | $\theta_i$ | $d_i$ | $a_i$ | $\alpha_i$ |
|---|------------|-------|-------|------------|
| 1 | $\theta_1$ | 0     | $a_1$ | 0          |
| 2 | $\theta_2$ | 0     | $a_2$ | 0          |
| 3 | $\theta_3$ | 0     | $a_3$ | 0          |

$$T_1^0 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & a_1 \cos \theta_1 \\ \sin \theta_1 & \cos \theta_1 & 0 & a_1 \sin \theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & a_2 \cos \theta_2 \\ \sin \theta_2 & \cos \theta_2 & 0 & a_2 \sin \theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^2 = \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & a_3 \cos \theta_3 \\ \sin \theta_3 & \cos \theta_3 & 0 & a_3 \sin \theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = T_1^0 * T_2^1 * T_3^2$$

$$T_2^0 = T_1^0 * T_2^1 = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = T_2^0 * T_3^2 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) + a_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2) + a_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) + a_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$y = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2) + a_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

Inverse kinematics

$$x_w = x - L_3 \cos \phi \therefore x_w = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y_w = y - L_3 \sin \phi \therefore y_w = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

Find  $\theta_2$

$$x_w^2 + y_w^2 = [L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)]^2 + [L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)]^2$$

$$x_w^2 + y_w^2 = [L_1^2 \cos^2 \theta_1 + 2L_1L_2 \cos \theta_1 \cos(\theta_1 + \theta_2) + L_2^2 \cos^2(\theta_1 + \theta_2)] + [L_1^2 \sin^2 \theta_1 + 2L_1L_2 \sin \theta_1 \sin(\theta_1 + \theta_2) + L_2^2 \sin^2(\theta_1 + \theta_2)]$$

$$x_w^2 + y_w^2 = L_1^2(\cos^2 \theta_1 + \sin^2 \theta_1) + L_2^2(\cos^2(\theta_1 + \theta_2) + \sin^2(\theta_1 + \theta_2)) + 2L_1L_2(\cos \theta_1 \cos(\theta_1 + \theta_2) + \sin \theta_1 \sin(\theta_1 + \theta_2))$$

$$x_w^2 + y_w^2 = L_1^2 + L_2^2 + 2L_1L_2 \cos \theta_2$$

$$\cos \theta_2 = \frac{x_w^2 + y_w^2 - L_1^2 - L_2^2}{2L_1L_2} \rightarrow \cos \theta_2 = A$$

$$\sin^2 \theta + \cos^2 \theta = 1 \therefore \sin^2 \theta + A^2 = 1$$

$$\sin \theta = \pm \sqrt{1 - A^2}$$

$$\theta_2 = \text{atan2} \left( \pm \sqrt{1 - A^2}, A \right) \therefore \begin{cases} \theta_{2A} = \text{atan2} \left( \sqrt{1 - A^2}, A \right) \\ \theta_{2B} = \text{atan2} \left( -\sqrt{1 - A^2}, A \right) \end{cases}$$

Find  $\theta_1$

$$x_w = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) \rightarrow L_1 \cos \theta_1 + L_2 \cos \theta_1 \cos \theta_2 - L_2 \sin \theta_1 \sin \theta_2$$

$$y_w = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) \rightarrow L_1 \sin \theta_1 + L_2 \sin \theta_1 \cos \theta_2 + L_2 \cos \theta_1 \sin \theta_2$$

$$x_w = (L_1 + L_2 \cos \theta_2) \cos \theta_1 - L_2 \sin \theta_2 \sin \theta_1$$

$$y_w = (L_1 + L_2 \cos \theta_2) \sin \theta_1 + L_2 \sin \theta_2 \cos \theta_1$$

$$\begin{aligned} A &= L_1 + L_2 \cos \theta_2 \\ B &= L_2 \sin \theta_2 \end{aligned}$$

$$\begin{aligned} x_w &= A \cos \theta_1 - B \sin \theta_1 \\ y_w &= A \sin \theta_1 + B \cos \theta_1 \end{aligned}$$

$$\begin{aligned} \begin{vmatrix} x_w \\ y_w \end{vmatrix} &= \begin{vmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{vmatrix} \begin{vmatrix} A \\ B \end{vmatrix} & \quad \begin{aligned} \beta &= \text{atan2}(B, A) \\ \alpha &= \text{atan2}(y_w, x_w) \end{aligned} \end{aligned}$$

$$\alpha = \theta_1 + \beta \therefore \theta_1 = \alpha - \beta$$

$$\theta_1 = \text{atan2}(y_w, x_w) - \text{atan2}(L_2 \sin \theta_2, L_1 + L_2 \cos \theta_2)$$

Find  $\theta_3$

$$\phi = \theta_1 + \theta_2 + \theta_3 \therefore \theta_3 = \phi - \theta_1 - \theta_2$$